Interface truncation and verified design gradients in a nonlinear motor model

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Abstract

A motor optimization target can change with the numerical interface even when average torque agrees closely. We reconstruct a published nonlinear permanent-magnet motor with piecewise-linear finite elements and compare conforming and harmonic rotor–stator coupling on identical meshes in our GEODE implementation. At the finest mesh, truncation at the reference's harmonic order 45 lowers loaded torque standard deviation by 7–9% and raises cogging standard deviation by 59%, relative to conforming coupling. Retaining every order below the trace Nyquist limit reproduces conforming torque within a few $10^{-9}\,\mathrm{N\,m}$ on the two checked meshes. Refining the independent reference reduces its cogging discrepancy with GEODE from 12.9% to 0.10%, although neither sequence is fully converged. For the saved optimized geometry, increasing interface order from 45 to 93 multiplies the aggregate squared ripple by 3.6 under fixed unscaled current, while the initial objective rises only 16%; the reduction remains above 2,400-fold across the tested models. We differentiate the converged nonlinear model through mesh motion, excitation and torque evaluation. Among 704 distinct loaded derivative comparisons, 688 agree with best-step central differences within 2.8×10^{-7} relatively; the largest error is 1.1×10^{-5}. Ripple sensitivities nevertheless differ between the verified model–estimator pairs. Fixing winding-group-1 density instead of current reverses three mean-torque sensitivities at the initial design on the second-coarsest mesh, at order 45 and full current. The results separate discrete derivative accuracy from model-dependent objectives and sensitivities; optimization time to a common validated target remains to be measured.

1 Introduction

A small error in mean torque can conceal a large change in the quantity being optimized. Reconstructing the initial motor of Wiesheu et al. (2024), we find that truncating the interface at order 45 lowers loaded torque standard deviation by 7–9% and raises cogging standard deviation by 59% on the finest mesh, relative to conforming coupling. The loaded mean torque of either model agrees with the saved reference within 0.16%. Varying the transmitted order and checking the full-order identity distinguish this model dependence from spatial refinement error. Rerunning and refining the independent reference accounts for the initial cogging discrepancy; checking the saved optimized geometry shows that interface order also affects the reported design improvement.

The same distinction applies to derivatives. An adjoint can produce many design sensitivities without a field solve per parameter (Scotzniovsky et al., 2025), but it differentiates the chosen discrete model and excitation policy. It does not establish that the objective is resolved, or that following its gradients beats a cheaper design model. These questions require separate checks of the field model, derivatives and optimization outcome.

We report a four-mesh comparison of conforming and truncated harmonic coupling, including two torque estimators and a harmonic-order ladder. We also verify derivatives of mean torque and

ripple with respect to 12 design parameters at initial and perturbed designs, and expose the effect of holding current rather than current density fixed. The comparison shows how interface truncation and excitation policy affect the objective and its sensitivities even when the discrete derivatives pass verification. The numerical methods are established; the results supply the evidence needed before comparing optimization costs.

2 Related work

Wiesheu et al. (2024) combine spline-based rotor and stator shape optimization with engineering parameters for a nonlinear permanent-magnet synchronous motor. Their public implementation and saved outputs provide the reproduction target here. Their related study combines parameter and shape optimization with isogeometric analysis and compares against each alone (Wiesheu et al., 2025). We inherit the initial geometry, material table and excitation conventions; we replace the field discretization and compare two rotor–stator couplings. The resulting comparison is between numerical models, rather than a comparison with a measured physical motor.

Scotzniovsky et al. (2025) combine free-form deformation, differentiable mesh warping and electromagnetic analysis with adjoint-based derivatives. Merkel et al. (2021) use isogeometric shape derivatives and harmonic rotor–stator coupling to reduce electromotive-force harmonic distortion. Their objective is electromotive-force distortion, and their validation compares initial and optimized designs with a separate field solver. Here we examine torque standard deviation in a nonlinear reconstruction and test which interface model and excitation policy its gradients represent.

Harmonic weighting of a sliding interface was developed for finite-element machine models by De Gersem and Weiland (2004). Bontinck et al. (2018) use separate isogeometric subdomains and harmonic interface constraints; their exact circular geometry avoids polygonal air-gap error. Egger et al. (2022) derive torque from the energy of a harmonic mortar formulation and examine its numerical accuracy for ripple evaluation. Their linear-material cogging experiment varies the maximum transmitted order from 39 to 129 and compares against a refined order-200 reference. Thus torque sensitivity to interface order is already established; our study quantifies its effect on loaded nonlinear ripple, a saved optimized design, and design sensitivities. Bontinck et al. also analyze stability as the multiplier space grows relative to the field space, so adding orders requires compatible spatial resolution. We compare coupling torque with the gap-averaged stress expression of Arkkio (1987). These works motivate checking interface order, field resolution and torque evaluation together, rather than treating small mean-torque error as sufficient validation. The sensitivity of cogging-torque prediction to finite-element discretization and evaluation method was examined by Howe and Zhu (1992); Arkkio and Hannukainen (2012) study mesh selection for torque computation.

Ion et al. (2018) compare deterministic gradient-based optimization with stochastic search for electric devices, including a permanent-magnet motor, using geometry parametrizations that avoid remeshing. Extending that comparison here requires competitive inexpensive models and evaluation of all candidates against a common refined target.

Kemmetmüller et al. (2015) use nonlinear magnetic equivalent circuits for torque control, including saturation and nonsinusoidal behavior. Their control problem differs from shape optimization, but it demonstrates why saturation alone does not establish the need for a full-field model. Farshbaf Roomi et al. (2022) develop a waveform-targeting surrogate for electrical machine design, while Wen and Zahr (2023) adapt reduced models along a shape-optimization trajectory in fluid examples. These approaches motivate waveform-aware and adaptive baselines. Their published performance is not transferred to the motor studied here.

3 Motor model and reconstruction

3.1 Geometry and operating states

The full machine has six poles and 36 stator slots. We model a 60° sector, occupying $60° \le \theta \le 120°$ at zero rotation, with antiperiodic conditions on the radial boundaries. The rotor inner and outer radii are 16 and 44 mm; the stator inner and outer radii are 45 and 67.5 mm. Axial length is $L = 0.1$ m. The 1 mm air gap is split at the coupling radius $r_c = 44.5$ mm. Figure 1 shows the material-tagged reconstruction.

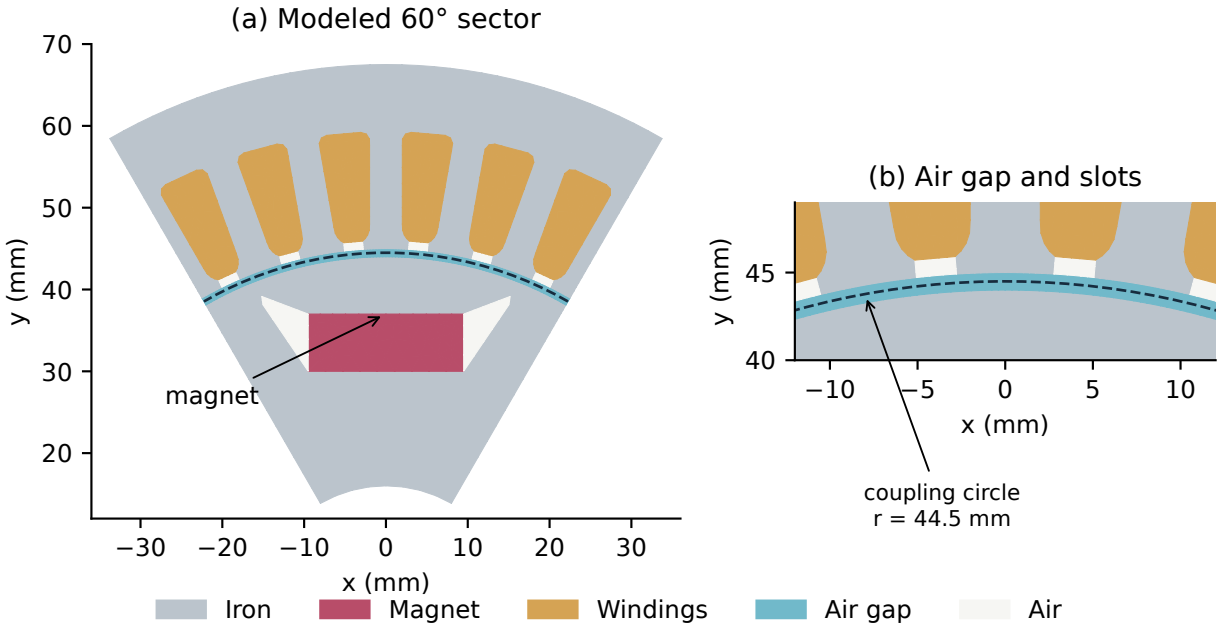


Figure 1: Initial-design geometry rendered directly from the committed coarsest mesh. The dashed line is the rotor–stator coupling circle. The right panel enlarges the gap and slot openings. Material boundaries are shown without the element edges; the geometry is not an optimized result.

The reconstruction uses the initial NURBS exports in the public snapshot. Their control points are stored with weights already multiplied in, and patch orientation must be accounted for when integrating areas. Two thin rotor patches described as air slits in source comments receive iron under the reference's material assignment; we follow that assignment. The mesh generator preserves the winding groups separately, since their two characteristic areas are approximately 36.29 and 31.22 mm^2.

At mechanical angle β, phase currents are

$$I_q(\beta) = aI_* \sin(\phi_0 + 3\beta + \delta_q), \qquad (\delta_U, \delta_V, \delta_W) = (0, -2\pi/3, 2\pi/3), \tag{1}$$

where $a \in \{0, 0.5, 1\}$, $I_* = 9.677657374$ A, and the electrical phase offset $\phi_0 = 0$ for the initial design. Angles in Eq. (1) are in radians. Each winding group carries current density $J_g = 12 s_g I_q / |\Omega_g|$, with its own area and prescribed sign $s_g \in \{-1, 1\}$. Using a common area would change the current density between groups by about 16%. We compare 20 distinct mechanical angles, $0°, 1°, \ldots, 19°$, excluding the duplicated endpoint in the source CSV. The reference's optimization grid contains the ten even angles; both grids are retained in the artifacts.

3.2 Nonlinear field equation

The two-dimensional magnetic vector potential $A = A_z$ determines the flux density $\mathbf{B} = (\partial_y A, -\partial_x A)$. Iron is described by an isotropic reluctivity $\nu(|\mathbf{B}|^2)$, where reluctivity is the inverse of permeability. For a test function v that satisfies the homogeneous constraints, the weak equation is

$$\int_\Omega \nu(|\nabla A|^2)\nabla A \cdot \nabla v \,\mathrm{d}\Omega = \int_\Omega J_z v \,\mathrm{d}\Omega + \int_{\Omega_m} \mathbf{M} \cdot (\partial_y v, -\partial_x v)\,\mathrm{d}\Omega. \tag{2}$$

Here $\mathbf{M}$ denotes the permanent-magnet source in $\mathbf{H} = \nu\mathbf{B} - \mathbf{M}$. In the magnet, $\mu_r = 1.05$ and $B_r = 1.4\,\mathrm{T}$, so its source magnitude is $H_c = B_r/(\mu_0\mu_r)$, not B_r/μ_0. Its direction is $\pi/2$ in the rotor frame and rotates with the rotor. Air and windings have $\mu_r = 1$. The inner and outer circular boundaries have $A = 0$.

We use continuous piecewise-linear triangles. With nodal vector u, element gradients $g_i = \nabla\varphi_i$, $G = \nabla A$ and $s = G \cdot G$, the residual derivative on one element is

$$K^e_{ij} = |\Omega_e| \left[\nu(s) g_i \cdot g_j + 2\nu'(s)(g_i \cdot G)(g_j \cdot G)\right]. \tag{3}$$

The second term is required when the material changes with field strength. Newton steps use this consistent tangent, with damping and a stopping reference independent of the initial guess. Constant potential offsets, where allowed by the boundary conditions, do not change that reference. The solver reports whether it stopped at the requested tolerance or an estimated floating-point floor. All 500 baseline and sensitivity solves, and all 1,140 additional harmonic-coupling solves, stopped at the requested tolerance.

The iron law uses 47 tabulated M27 B–H points from 0 to 2.30 T. Monotonicity is imposed on $H(B)$, not on $\nu(B^2) = H(B)/B$. The tabulated H/B is non-monotone up to about 0.75 T. The two material tangent eigenvalues must satisfy

$$\nu > 0, \qquad \nu + 2s\nu' = \frac{\mathrm{d}H}{\mathrm{d}B} > 0. \tag{4}$$

The GEODE and reference interpolants agree to round-off between 0.05 and 2.25 T. GEODE uses an odd cubic near zero and a continuous slope into the vacuum-slope tail above 2.30 T. The reference uses a low-field clamp and a different last-knot slope. The maximum relative difference in reluctivity near zero is 6.66%; the last segment differs in H by at most 2.17×10^{-4} relatively. We test the torque effect of the low-field difference in Section 5.3.

3.3 Rotation and torque

At the modeled angles, the two circular traces have matching nodes after rotation. Our conforming interface identifies their potentials, including the antiperiodic sign on sector wrap-around. Writing the full nodal vector as $u = P\hat{u} + u_D$, the reduced residual and tangent are $\hat{R} = P^T R(P\hat{u} + u_D)$ and $\hat{K} = P^T K P$. The rotor geometry and magnet source rotate together; currents use the same mechanical angle in Eq. (1).

We average Maxwell stress over the air gap using the Arkkio expression (Arkkio, 1987)

$$T(\beta) = \frac{L}{\mu_0(r_2 - r_1)} \int_{\Omega_{\mathrm{gap}}} r B_r B_\theta \,\mathrm{d}\Omega, \qquad (r_1, r_2) = (44, 45)\ \mathrm{mm}. \tag{5}$$

Positive torque acts counter-clockwise on the rotor. All torques in this paper include axial length and refer to the modeled sector. We apply no factor of six when comparing with the saved reference arrays, despite a per-length label appearing in the reference evaluation plots.

The alternative harmonic model imposes

$$\int_{\Gamma_c} (A_s - A_r) e_n \, \mathrm{d}\Gamma = 0, \qquad e_n \in \{\sin(n\theta), \cos(n\theta)\}, \quad n = 3, 9, \ldots, 45. \tag{6}$$

The 16 constraints match the reference's eight orders. Rotor and stator retain distinct trace unknowns. With constraint matrix C, Lagrange multipliers η and reduced potential $\hat{u}$, Newton's method uses the saddle tangent

$$\begin{bmatrix} K & C \\ C^T & 0 \end{bmatrix}. \tag{7}$$

The trace moments use the exact Fourier integrals of the piecewise-linear hats. Rotation enters their centers; the constrained energy derivative gives

$$T_c = -L\eta^T \frac{\partial (C^T \hat{u})}{\partial \beta}, \tag{8}$$

with the state held fixed in the partial derivative, consistent with Egger et al. (2022). Forward comparisons use Arkkio for both GEODE couplings to isolate the interface change; coupling torque is a separate check. The derivative study uses coupling torque for the harmonic model and Arkkio for the conforming model. The two objectives and their gradients are consequently comparisons of those specified model–estimator pairs.

The main harmonic runs retain antiperiodic trace-end pairing. An additional set frees those ends, matching the reference's treatment. Increasing the transmitted orders to every mode below the trace Nyquist limit reproduces the conforming Arkkio torques within $4.1 \times 10^{-9}\,\mathrm{N\,m}$ on L0 and $1.3 \times 10^{-9}\,\mathrm{N\,m}$ on L1. This checks the interface implementation without fitting any parameter to the saved torque.

4 Verification and experiment design

The nonlinear implementation is checked against linear assembly, residual finite differences, a manufactured solution and a saturated annulus with an independent radial field relation. Solver verification records are linked in the data-availability section; motor derivative checks appear in Section 6.

Sector tests compare the constrained solution with a full-machine solve, check the reduced tangent by finite differences, and exercise positive, negative and wrapped rotation steps. Clockwise and counter-clockwise triangle orderings give equivalent magnet sources and fields. The motor regression also detects changes in magnet strength, magnet direction, phase ordering and rotation direction.

The motor study uses four meshes, L0–L3, with 3,153, 9,937, 36,083 and 139,268 nodes. The 60° coupling arc has $120 \cdot 2^{\ell}$ segments at level ℓ, so integer-degree rotation is represented at every level. Each level contains 60 baseline solves: three current levels and 20 angles. A solution at one angle initializes the next angle at the same current. We report the mean and population standard deviation,

$$\bar{T} = \frac{1}{N} \sum_{k=1}^{N} T_k, \qquad \sigma_T = \left[\frac{1}{N} \sum_{k=1}^{N} (T_k - \bar{T})^2 \right]^{1/2}, \quad N = 20. \tag{9}$$

The standard deviation is the ripple metric used for the comparison. At zero current it measures cogging, the torque variation caused by the permanent magnets and slots without applied phase current.

For a scalar statistic q, the estimated L3 refinement interval is

$$q_3 \pm e_q, \qquad e_q = \max\{|q_2 - q_1|, |q_3 - q_2|\}. \tag{10}$$

We use the same construction pointwise for waveforms. This interval is a refinement diagnostic, not a proven error bound or a statistical confidence interval. It does not cover reference-model error or angular quadrature error. The loaded means change direction between the last two increments, so a last-increment estimate alone would be particularly weak.

5 Forward reproduction results

5.1 Coupling changes the ripple objective

Table 1 compares the same geometry, material and excitation with two interface models. The conforming L3 means are 0.15% and 0.16% low at half and full current. Harmonic coupling reduces the mean deviations to −0.002% and +0.051%, with both reference means inside the estimated refinement intervals. The loaded ripple deviations fall from +9.9% and +11.3% to +1.7% and +1.3%.

Table 1: L3 population standard deviation and mean torque, in N m per sector, on the 20-angle grid. The last column gives ripple deviation from the saved reference; n/a identifies the reference itself. The reference is subsequently refined in Section 5.4.

Current fraction	Model	Mean	Std	Std vs ref. (%)
0	Reference	0.00010	0.01071	n/a
0	Conforming	0.00001	0.00760	-29.1
0	Harmonic	0.00000	0.01209	+12.9
0.5	Reference	0.66208	0.03895	n/a
0.5	Conforming	0.66106	0.04279	+9.9
0.5	Harmonic	0.66206	0.03961	+1.7
1	Reference	1.44248	0.07935	n/a
1	Conforming	1.44023	0.08829	+11.3
1	Harmonic	1.44322	0.08036	+1.3

Figure 2 shows the improvement angle by angle. The largest full-current difference falls from 0.0272 to 0.0034 N m. The harmonic full-current ripple interval contains the saved reference; the saved half-current value lies approximately 5×10^{-5} N m below its interval. Mean agreement is therefore insufficient to characterize waveform accuracy.

The saved zero-current reference exposes a separate discretization effect. Harmonic coupling changes the cogging standard deviation from 29.1% below to 12.9% above the reference. Its waveform is close to a 1.13-fold scaling of the saved waveform, but the reference lies outside the estimated ripple interval. This motivates refining the reference itself before interpreting the discrepancy. The harmonic cogging standard deviations are 0.0101, 0.0119, 0.0121 and 0.0121 N m on L0–L3, compared with 0.0107 N m in the reference. The independent reference refinement below tests this coarse-model explanation directly.

5.2 Spatial resolution and transmitted order

Spatial refinement at fixed order and order refinement on a fixed mesh answer different questions (Fig. 3). At L3, order 45 raises cogging standard deviation by about 59% and lowers loaded standard deviation by 7–9%, relative to the conforming values. At L1 and L2, order 189 approaches the conforming torque to within 1.1×10^{-4} N m. A spatially refined order-45 solution thus remains a

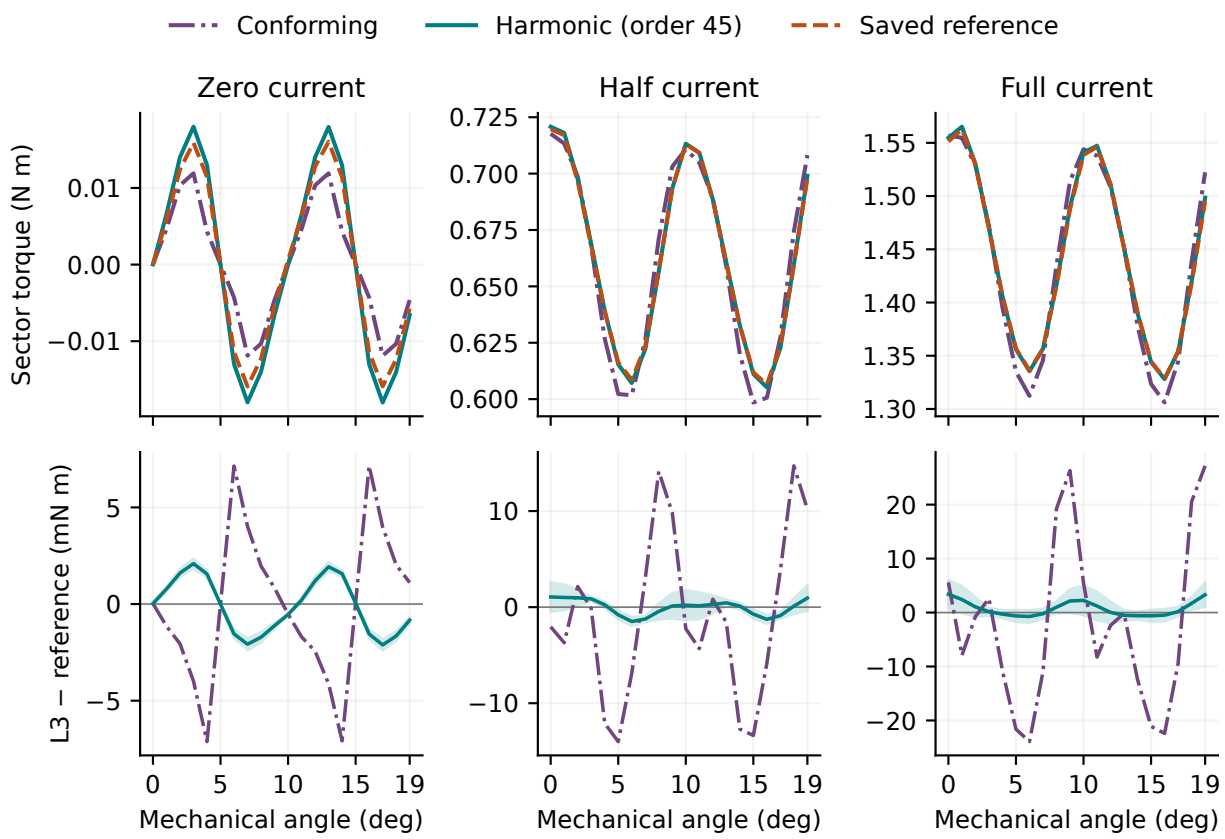


Figure 2: Top: L3 torque for conforming and order-45 harmonic coupling, compared with the saved reference. Bottom: each GEODE waveform minus the reference; shading is the harmonic model's pointwise estimated refinement interval from Eq. (10). Residual panels have independent scales. All GEODE curves use Arkkio torque, so the paired difference changes the coupling while holding the estimator fixed. The saved reference is coarse; Section 5.4 reports its refinement.

solution of a truncated interface model. Matching that truncation improves reproduction of the published calculation; it does not establish which model better represents the physical motor.

Two additional controls narrow the explanation. At L3, coupling torque and the inner- and outer-half Arkkio estimates are each within 9×10^{-5} N m of full-band Arkkio in the order-45 solution. Freeing the coupling-arc ends changes torque by less than 5.4×10^{-5} N m. These effects are too small to explain the remaining cogging gap at the tested resolution.

5.3 Sensitivity controls

The controlled changes in Table 2 are too small to explain the observed waveform gap at the tested settings. The near-zero material difference has particularly little effect on torque despite its 6.66% local reluctivity difference. This is a measured sensitivity of the initial design and does not establish insensitivity after optimization.

In the recorded estimator study, single-circle Maxwell-stress estimates vary more with the integration radius than the half-band Arkkio estimates. The paired coupling-torque and half-band checks above avoid selecting a favorable radius. A post-processing filter of the solved conforming field also gives non-monotone ripple changes across the tested cuts and currents; it does not replace a field solve with a truncated interface. The complete filter results are retained in the accompanying

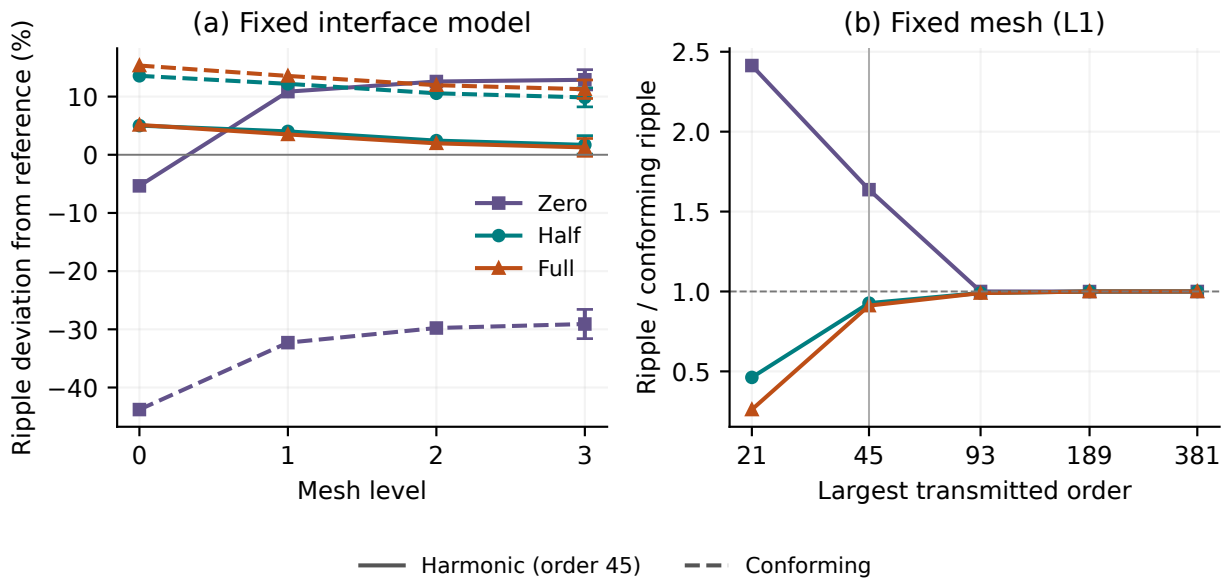


Figure 3: Left: ripple deviation from the reference under spatial refinement, with solid lines for harmonic order 45 and dashed lines for conforming coupling; L3 bars use Eq. (10). Right: order refinement on L1 with logarithmic order spacing, normalized by the conforming ripple at that same level. The saved reference uses order 45; see Section 5.4 for its refinement. Colors identify current, not coupling.

Table 2: Maximum absolute torque change relative to the baseline at the same mesh level, using conforming coupling and Arkkio torque. Entries summarize the recorded angle/current subsets.

Change	Levels	Max. $\lvert\Delta T\rvert$ (N m)
Reference low-field interpolation and clamp	L2, L3	2.1×10^{-6}
Relative Newton tolerance 10^{-12} vs 10^{-8}	L2; L3 full current	7.3×10^{-8}
Exact winding areas vs mesh areas	L2	3.9×10^{-4}
Inner or outer half of the Arkkio band	L3	7.3×10^{-5}

forward-study records.

5.4 Independent reference refinement

We rerun the public reference with its own degree-two isogeometric model, harmonic orders $3, 9, \ldots, 45$ and coupling torque, at 3, 6, 12 and 24 subdivisions in each patch direction. The 3×3 rerun first reproduces all 60 saved torques with differences at most 9.8×10^{-13} N m. The run uses Octave 11.3.0, GeoPDEs revision `53ae5ca1` and nurbs 1.4.3; compatibility edits, source hashes and raw records are retained. The corresponding field degrees of freedom are 4,079, 12,518, 43,220 and 159,920.

The reference cogging standard deviation changes from 0.01071 to 0.01220, 0.01212 and 0.01208 N m (Fig. 4a). GEODE L3 at order 45 gives 0.01209 N m, a 0.10% difference from the finest reference. The maximum pointwise discrepancy drops from 2.1×10^{-3} to 3.4×10^{-5} N m. At half and full current, the finest reference standard deviations are 0.03947 and 0.08014 N m; GEODE is 0.35% and 0.27% higher (panel b). All five finest-reference statistics, the three standard deviations and two loaded means, lie within GEODE's L3 refinement intervals. The coarse saved reference therefore accounts for the cogging mismatch in Section 5.1 at the matched interface order.

This agreement still concerns finite discretizations. The last cogging increments approximately

halve; extending that trend would put a limit near 0.0120 N m, an estimate without a bound. The last two half-current standard-deviation increments are both about −0.3% and do not shrink; we therefore leave its remaining error unestimated. The 0.10% cogging agreement is a comparison of finite-resolution values, not an accuracy bound; neither refinement sequence is fully converged. Tightening the reference Newton stopping threshold from 10^{-6} to 10^{-14} changes the checked cogging torques by at most 4.4×10^{-8} N m. GEODE retains straight-sided geometry, and the high-field extrapolation of the iron law has not been varied in a paired torque experiment. Neither cross-code agreement nor refinement proves physical accuracy for the machine.

5.5 Validation of the saved optimized design

The saved reference design at iteration 120, denoted Opt120, includes 12 scalar parameters and ten grouped surface-control offsets. We evaluate its full exported geometry on freshly generated meshes; simply moving the finest initial mesh produces four inverted triangles. Its inner air-gap band no longer has the annular area assumed by the full-band Arkkio functional, so harmonic results here use coupling torque. Conforming results use only outer-band Arkkio on the fresh exported mesh. On the same L3, order-45 Opt120 field at full current, the outer-band and coupling estimates differ by about 1.3×10^{-4} N m in mean and 1.1×10^{-5} N m in standard deviation. This paired check supports the estimator substitution at this resolution. This estimator restriction is specific to these geometries; it does not validate outer-band Arkkio on arbitrary moved meshes or stator offsets.

Define $F = \sum_{a\in\{0,0.5,1\}} \sigma_T(a)^2$, in $\mathrm{N^2\,m^2}$, on the 20-angle grid. With the evaluation current held fixed in the unscaled geometry, raising the transmitted order from 45 to 93 multiplies Opt120's F by 3.6, from 1.07×10^{-6} to 3.86×10^{-6} (Fig. 4c). The initial design's F rises by 16%. Its ratio to Opt120 is consequently 7,631 at order 45, 2,454 at order 93 and 2,444 for conforming coupling. The full-current Opt120 standard deviation rises from 0.00066 to 0.00164 N m. The large ripple reduction survives these checks, but its magnitude depends strongly on the interface model. Between L2 and L3, changes in F, normalized by L3, reach 1.36% for the initial design and 2.71% for Opt120 across the three models; these increments are not error bounds. For Opt120 at order 45, the successive L1–L2 and L2–L3 increases in F are 3.05×10^{-8} and $2.90 \times 10^{-8}\,\mathrm{N^2\,m^2}$, so they barely shrink and do not establish an asymptotic rate. An alternate freshly constructed mesh family changes by 3.22% from L2 to L3. The $2.79 \times 10^{-6}\,\mathrm{N^2\,m^2}$ interface effect is about 96 times the last export-mesh increment, despite this unresolved remainder. A possible explanation is that optimization against order 45 favors cancellation within that truncated model. This is a conjecture, not an ablation result: increasing the order raises full-current ripple but lowers Opt120 cogging standard deviation from 0.00061 to 0.00035 N m.

Current policy also changes feasibility (panel d). At L3 and order 45, the initial full-current mean is 1.4433 N m. Opt120 gives 1.3147 N m under fixed *unscaled* current, 1.2550 N m under fixed physical current, and 1.4983 N m under the reference's fixed group-1 density rule. The first two are 8.9% and 13.0% below the initial mean; the density-rule value is 3.8% above it. These policy comparisons use the same Opt120 geometry, but comparisons with the initial motor also change size: Opt120 shortens the stack from 100 to 80 mm and has radial scale $s = 1.0477$, unscaled outer diameter 130.293 mm and operating angle −11.646° (Table 4). Every Opt120 torque carries the explicit size factor $s^2 L/(0.1\,\mathrm{m}) = 0.878175$. Removing both size factors from the fixed-unscaled-current mean gives 1.4971 N m, 3.7% above the initial mean; normalizing only length to 100 mm while retaining the actual radial scale gives 1.6434 N m. These algebraic normalizations separate the size factors from the current rule; they are not reoptimized designs. Opt120 is outside the prospective fixed-size, fixed-physical-current problem.

A separate reproduction comparison uses the saved reference current at each design, rather than fixed unscaled current. At Opt120, GEODE's order-45 full-current mean is 1.4986 N m versus

1.5000 N m in the saved arrays, but its standard deviation is 3.53×10^{-4} versus 1.76×10^{-4} N m, a factor of 2.0. The half-current values are 5.53×10^{-4} and 4.37×10^{-4} N m. The largest pointwise difference across the two loaded currents is about 2.1×10^{-3} N m. With this matched current policy and zero-current values included, the initial-to-Opt120 objective ratios are 10,170 for GEODE and 11,982 for the saved reference. The 7,631 ratio above instead holds unscaled current fixed. Reference refinement was run only at the initial design, so the cause of the Opt120 discrepancy remains unresolved.

Angular sampling becomes more consequential at this low ripple. Relative to the 40-angle grid, the ten-angle Opt120 objective reads about 1.05% high at order 45 and 5.90% low at order 93 on the fresh L3 mesh; the conforming value reads 17.1% low. The accompanying records retain shifted 20-angle grids and withheld currents 0.25 and 0.75. Candidate validation must check these effects at the candidate, since initial-design agreement alone would miss them. All 3,240 solve rows in the 18 successful records used for these candidate comparisons stopped at tolerance.

6 Design-gradient verification

6.1 Differentiated model and excitation policy

For a design vector p, the converged state z contains the reduced potential and, for harmonic coupling, its multipliers. Writing the full constrained residual as $R(z, p) = 0$, the discrete adjoint is

$$R_z^T \psi = F_z^T, \qquad \frac{\mathrm{d}F}{\mathrm{d}p} = F_p - \psi^T R_p. \tag{11}$$

We differentiate mesh coordinates, element terms, magnet sources, current loads, harmonic constraints and the torque functional. The explicit coordinate derivative of Arkkio is included for conforming coupling. Fixed-topology mesh motion $X(p)$ supplies the chain rule $\mathrm{d}F/\,\mathrm{d}p_k = \langle \mathrm{d}F/\,\mathrm{d}X, \partial X/\partial p_k \rangle$ for geometric parameters, with direct terms added where needed.

The 12 parameters comprise nine geometric lengths (MAG, MH, MW, SW1–SW4, SD1 and Sr1), axial length, operating angle and radial scale. The two designs are the initial parameters and an interior point halfway toward the saved optimized scalar parameters (Appendix B). This verifies a local scalar parametrization. The reference's optimized geometry also uses grouped surface-control displacements; reproducing its 12 scalar values alone does not reproduce that geometry. The present gradient comparison does not include derivatives of those additional groups or a new optimized motor.

In this scalar derivative study, fixed current means fixed peak current in the unscaled geometry, so $J_g = 12 s_g I_q / |\Omega_g(X)|$ changes when a slot changes area. For comparison we implement the reference optimizer's rule, denoted *fixed group-1 density*: $I_* \propto |\Omega_1(X)| s$, where s is radial scale, with the proportionality constant fixed at the initial design. The unscaled geometry is evaluated and torque is multiplied by s^2, following the reference. Both rules agree at the initial geometry but have different derivatives. Group-2 density is not held fixed because the ratio of the two winding areas can change. At L1 (harmonic coupling, coupling torque, full current), the mean-torque derivative with respect to SW1 is $+1.68$ N m/m under fixed current and -220.85 N m/m under fixed group-1 density (Fig. 5). SW1 is the stator-tooth-width parameter. The sign change comes from the additional winding-area/current term: a wider tooth narrows the slot and lowers current under fixed group-1 density. At this state the policy changes seven of the ten geometric/scale derivatives and reverses the mean-torque signs for SW1, SW4 and SD1. Rotor parameters MAG, MH and MW leave winding area unchanged and therefore give identical derivatives under the two rules. The prospective fixed-size problem sets radial scale to one, making fixed physical and unscaled current coincide. A future optimization comparison must hold this policy fixed across methods.

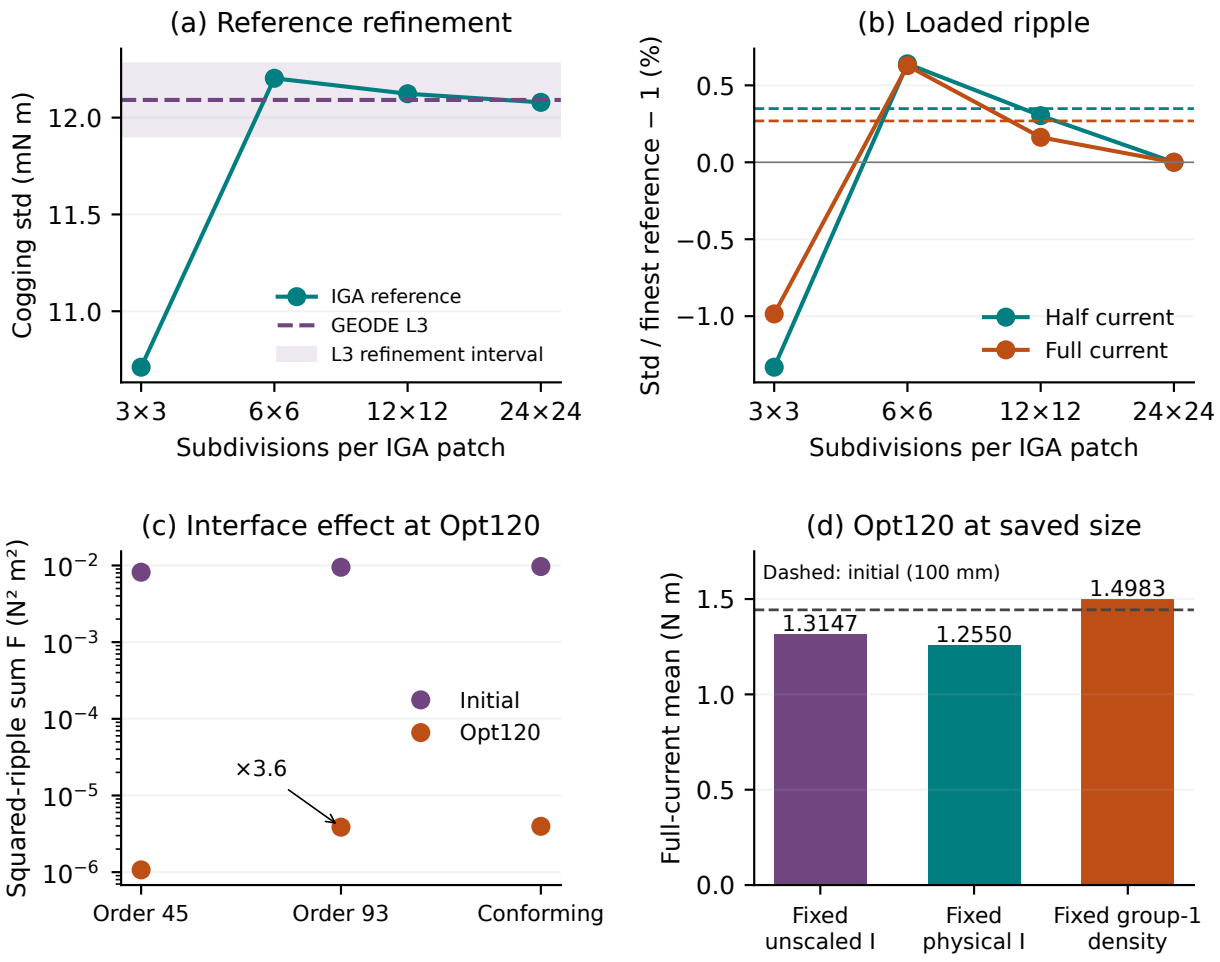


Figure 4: Reference and candidate validation from raw torque records. (a) Independent IGA reference cogging refinement at order 45; the dashed line is GEODE L3 and its shaded band uses Eq. (10). (b) Loaded standard deviation relative to the finest reference at each current; circles are IGA, horizontal lines are GEODE L3. The finest IGA point is zero by construction. Panels (a,b) use the initial geometry and 20 angles; GEODE uses Arkkio, with coupling torque checked separately. (c) Initial and Opt120 squared ripple sums at L3 on fresh meshes and 20 angles, under fixed unscaled current. Harmonic values use coupling torque; conforming values use outer-band Arkkio. (d) Full-current mean at order 45 for Opt120 under three current rules at its saved size; the dashed line is the initial mean at the initial size. All candidate values use coupling torque.

6.2 Finite differences and Taylor remainders

The derivative objectives are mean torque and population standard deviation on the ten even mechanical angles, matching the reference's optimization grid. Central differences move the same mesh at $p \pm he_k$; step sweeps separate truncation from floating-point and nonlinear-solve error. We also check the first-order remainder $|F(p + he_k) - F(p) - h\ \mathrm{d}F/\mathrm{d}p_k|$ for quadratic decay.

Across initial and perturbed designs, L0–L2 and the recorded loaded current/rule subsets, there are 800 objective–parameter records. They pool harmonic/coupling-torque and conforming/Arkkio evaluations. The fixed-density MAG, MH and MW records repeat 96 fixed-current comparisons byte-for-byte, leaving 704 distinct comparisons. Of these, 688 have a best-step relative error at most 2.8×10^{-7}; the maximum is 1.07×10^{-5} for a small Sr1 ripple derivative. This is a best-step verification statistic, not the accuracy of a finite-difference step chosen in advance. Figure 5 includes

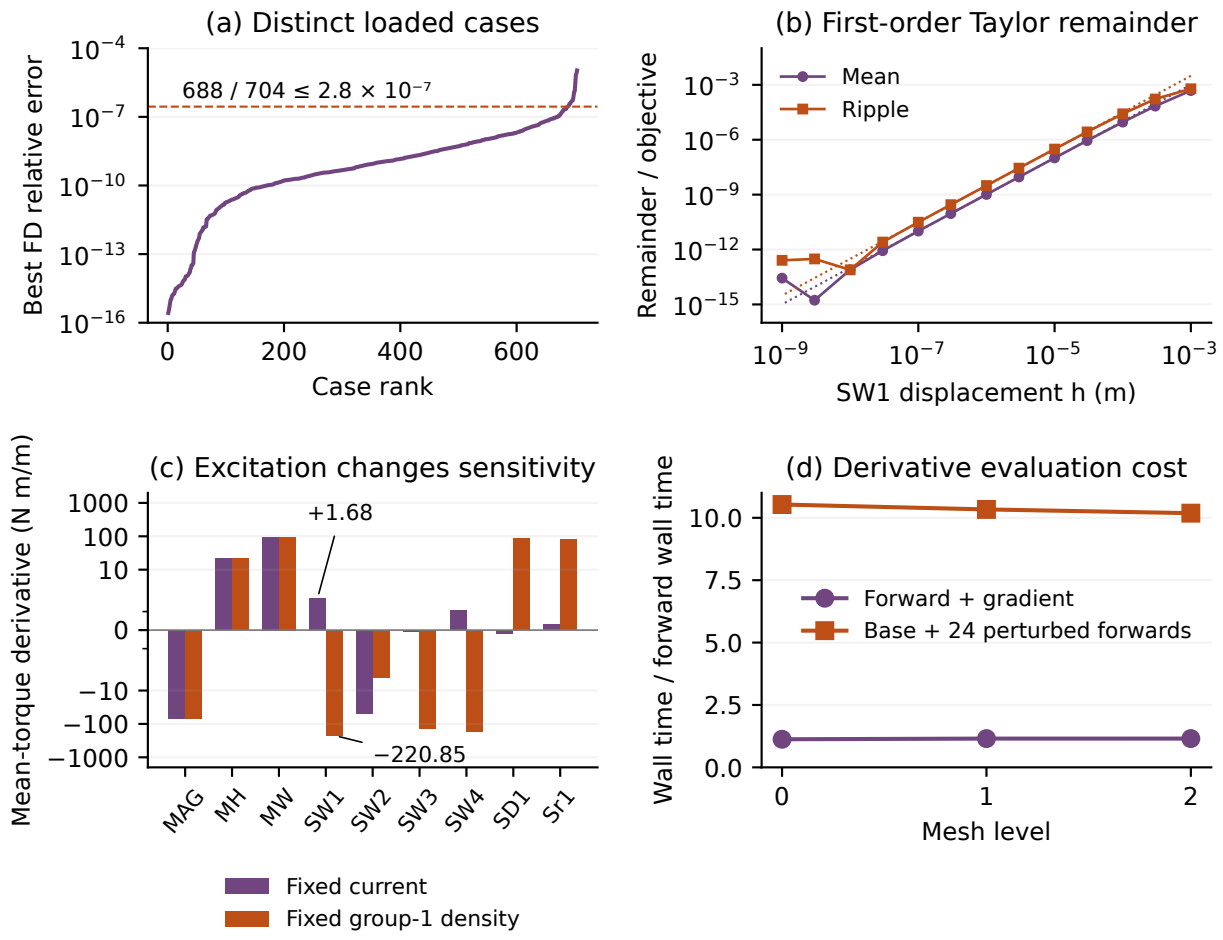


Figure 5: Discrete gradient evidence. (a) The 704 distinct loaded best-step finite-difference errors, sorted, pooling harmonic/coupling-torque and conforming/Arkkio objective–parameter cases. (b) SW1 Taylor remainders at L1, initial design, full current, harmonic coupling and coupling torque, fixed current; the plotted remainders are divided by $|F(p)|$, and dotted h^2 guides are anchored at $h = 10^{-5}$ m. (c) Mean-torque sensitivities for the nine geometric parameters under two current rules at that same state; the vertical axis is symmetric-log, linear within ± 2 N m/m. (d) Recorded cost ratios for harmonic coupling at tolerance 10^{-14}; gradients include the forward evaluation, and FD includes its shared base forward. L0/L1 use medians of three runs, L2 a single run. These timing data were collected under uncontrolled host load.

the 704 distinct cases and an explicit Taylor sweep. The artifact retains every step, rejected displacement and relative error. The recorded subsets do not form a full factorial study.

After removing duplicates and the analytic axial-length and radial-scale cases, 530 of 592 loaded comparisons have an anchored quadratic-remainder window spanning at least four decades in step size; 388 span at least five. The window is the contiguous run where remainder$/h^2$ stays within a factor of two of its value at $h = 10^{-5}$ m (or 10^{-2} degrees for operating angle). Smaller gradients reach round-off sooner. An anchored-window diagnostic fails for the initial-design, full-current conforming/fixed-density SW1 ripple cases at L0 and L1 because the remainder changes sign near the anchor; their widest quadratic windows span two decades. The nearly cancelling zero-current mean is a weaker check: its best relative errors reach 7.0×10^{-4}. Cogging standard-deviation errors are at most 8.9×10^{-7}. Thus the loaded agreement does not describe every zero-current derivative.

Discrete derivative accuracy also does not imply mesh convergence. In the harmonic model, SW3 mean-torque sensitivities change sign between coarse levels and their increments shrink with refinement; ripple gradients converge more slowly than most mean gradients. The conforming/Arkkio SW2 mean sensitivity is non-monotone across L0–L3 (-46.5, -41.3, -44.4, $-47.2\,\mathrm{N\,m/m}$), whereas harmonic/coupling torque approaches -47.1 from $-49.8\,\mathrm{N\,m/m}$. Fresh-remesh checks of MW, SW1 and SW2 are limited at small steps by changes in connectivity. The L2 MW mean-derivative fit has relative gap 2.6×10^{-6}, but individual remeshed steps at $h \geq 10^{-4}\,\mathrm{m}$ scatter by about 6×10^{-4}. The largest L2 full-current remeshed-versus-moved fitted discrepancy is 3.8% for the SW1 ripple derivative. These checks support the discrete chain rule while leaving refinement of candidate designs necessary.

6.3 Model dependence of verified sensitivities

The two model–estimator pairs give different ripple sensitivities even at the same design (Fig. 6). At L2 and full fixed current, the MAG ripple derivative is $1.28\,\mathrm{N\,m/m}$ for harmonic/coupling torque versus 1.94 for conforming/Arkkio at the initial design. At the perturbed design the values are 0.75 and $1.87\,\mathrm{N\,m/m}$. These differences persist under mesh refinement. At L3 the MAG pairs are 1.22 versus 1.87 at the initial design and 0.57 versus $1.67\,\mathrm{N\,m/m}$ at the perturbed design. In particular, the latter harmonic component changes appreciably even though the vector norm changes by about 1%. Mean-torque sensitivities agree more closely in the box-scaled comparison: the Euclidean norm of the vector difference, divided by the conforming vector norm, is 0.58% and 0.66% for the mean at the initial and perturbed designs, versus 6.7% and 10.2% for ripple. The norm weights the largest components most heavily: the initial SW2 and SW3 mean components differ by 6.5% and 17% relative to conforming values. At L3, the ripple gaps remain 6.58% and 10.25%, while each model changes by 0.46–1.19% from L2 to L3. The mean gaps fall to 0.16% and 0.26%; the conforming model itself changes by 0.52–0.54%. The L2 mean gap is therefore comparable to spatial refinement, whereas the ripple gap persists beyond the measured refinement increments. Because the estimator changes with the model, this experiment measures their combined effect; it does not isolate an interface-only derivative error or establish which gradient is closer to the physical machine.

6.4 Recorded derivative cost

The recorded timings use an Apple M3 Ultra with 28 cores, default solver threading, and concurrent host load averaging 14–24. The effective thread count was not pinned. A forward evaluation covers ten angles, starts from zero, and warm-starts successive angles. The full gradient includes that forward, ten torque adjoints, mesh derivatives and contractions for all 12 parameters and both aggregate objectives. Both aggregates reuse the per-angle torque derivatives. Central differences use 24 forward sweeps, each warm-started from the base states, at step $h = 10^{-6}$ in each parameter's stored unit (m, degrees or dimensionless scale). We add the shared base forward to the recorded 24-sweep FD time for a common starting point. These derivative timings are a separate workload from the 1,640 forward validation solves.

Across L0–L2, both couplings and tolerances 10^{-8} and 10^{-14}, the full gradient costs 1.13–1.20 forwards, whereas central differences cost 8.2–10.5 including the shared base solve. Table 3 gives the harmonic tight-tolerance subset. Shape-map setup is recorded separately and must be included in a complete optimization budget. The cost ratio concerns derivative evaluation on one loaded host. It provides no measurement of optimizer iterations, surrogate calibration cost or time to a validated target.

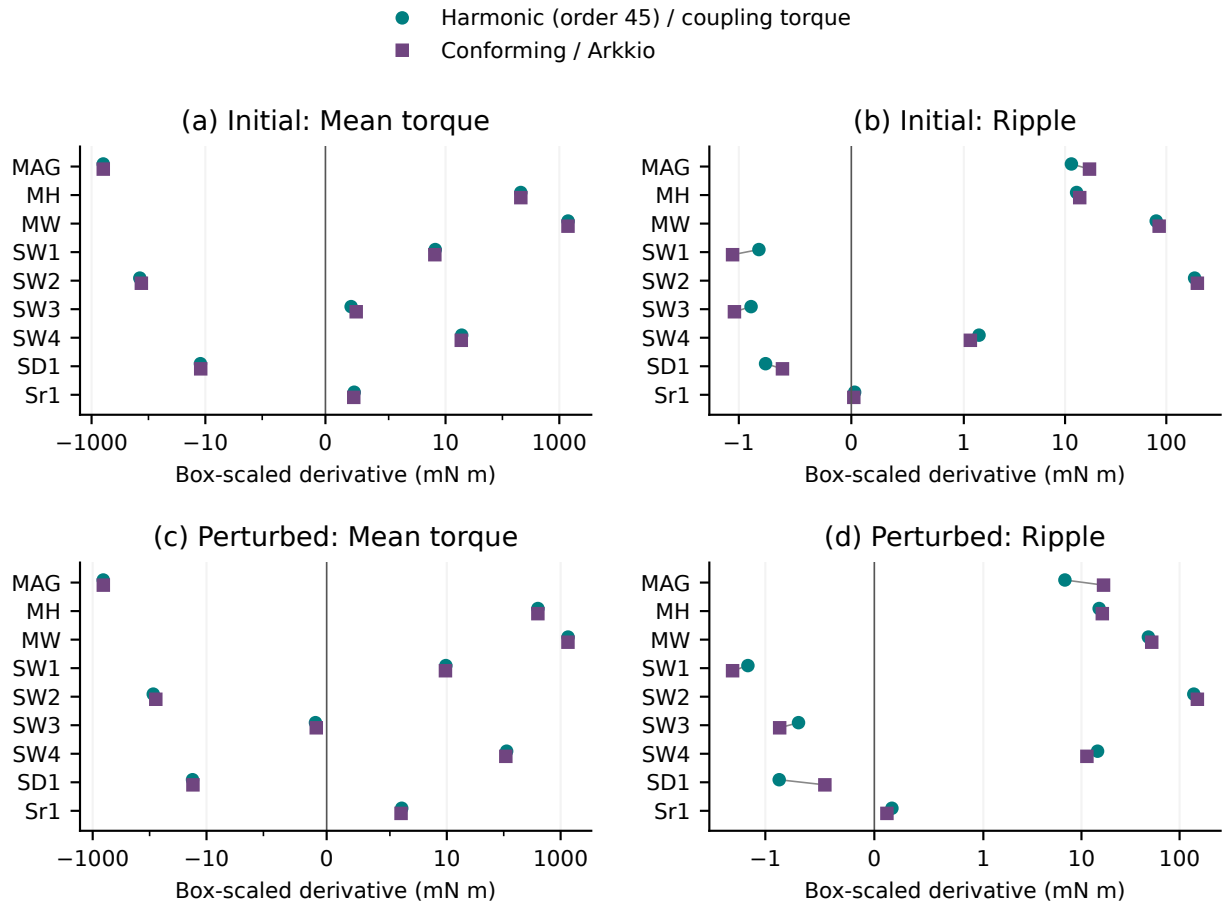


Figure 6: Full fixed-current sensitivities at L2, scaled by each geometric parameter's reference box width w_k. Top: initial design; bottom: perturbed design. L3 MAG values and refinement changes are given in the text. Left: mean torque; right: standard deviation. The plotted quantity $w_k \partial q / \partial p_k$ is a local sensitivity, not a finite change across the box. Coupling and torque estimator both differ between the two series. Gray segments join the two estimates for each parameter. Horizontal axes are symmetric-log, linear within ± 1 mN m. All nine widths, in mm, are 9, 10, 15, 4, 3, 1, 18, 70 and 1.5 in the displayed order.

Table 3: Recorded harmonic-model cost at relative tolerance 10^{-14}, initial design and full current. Times are seconds for ten-angle sweeps; FD includes the shared base sweep, and setup is a one-time shape-map cost.

Mesh	Forward	Forward + gradient	Base + FD	Setup
L0	0.664	0.748	6.997	0.011
L1	3.174	3.658	32.811	0.041
L2	12.702	14.644	129.390	0.222

7 Discussion and limitations

The controlled interface change explains most of the loaded ripple gap while preserving mean-torque agreement. The order ladder shows why this matters for optimization: interface truncation changes the objective even as spatial error decreases. The paired gradient comparison also measures changes in ripple sensitivities despite small discrete derivative errors. The saved optimized design retains a large ripple reduction after the checks reported here, while its objective increases 3.6-fold with

interface order and its mean-torque comparison depends on both motor size and current policy.

The model is two-dimensional and magnetostatic. It does not evaluate thermal behavior, demagnetization, fatigue, end effects, losses or a complete efficiency map. The comparison uses published computational artifacts, with no new measurements on a physical motor. The forward comparisons cover the initial and saved optimized designs; local derivative tests at an interior scalar design do not validate an entire optimization trajectory. The reference refinement is at the initial design. Denser and shifted angle grids and additional currents are checked for Opt120, but new candidates require their own refinement.

The prospective optimization comparison fixes outer diameter (SD1), axial length, radial scale, operating angle and physical peak current, and compares eight geometric scalars with those scalars plus ten grouped offsets. Exact-gradient, derivative-free and empirical radial-basis-surrogate methods will be assessed against predeclared 50%, 90% and 99% ripple-reduction targets, using a common order-93, L2, 20-angle validation with numerical and mean-torque margins and final L3, 40-angle checks. All costs count toward a budget of 60 measured validation-forward wall times; unattained targets remain censored. There are no matched optimization results in this paper, and an empirical surrogate alone cannot establish an advantage over a reduced physical model.

A cheap waveform or reduced physical model is an essential baseline. The linear slotless screening control in Appendix A admits a three-coefficient angular fit at one geometry. That does not establish generalization across shapes, but it cautions against inferring an optimization advantage from parameter count or adjoint cost alone.

8 Conclusion

On identical meshes of a published nonlinear motor, order-45 truncation lowers loaded torque standard deviation by 7–9% and raises cogging standard deviation by 59% relative to conforming coupling at L3. Refining the independent reference reduces the initial cogging discrepancy from 12.9% to 0.10% at matched order, although neither sequence is fully converged and the remaining error is not bounded. For the saved optimized design, raising the interface order from 45 to 93 multiplies the squared-ripple objective by 3.6 while preserving a reduction above 2,400-fold. Its shorter stack and changed radial scale must be separated from current policy when comparing mean torque with the initial motor; at its saved size it exceeds the initial mean only under the density rule among the three policies tested. Design gradients pass finite-difference and Taylor checks at two parameter points, while the cross-model ripple-sensitivity gap persists under refinement. The excitation rule reverses three mean sensitivities at the initial L1, order-45, full-current state. Derivative verification therefore does not settle objective resolution or feasibility. A practical speedup still requires optimization against inexpensive baselines at a common independently refined target.

Code and data availability

The original forward and gradient evidence is fixed at GEODE commit `910593ba`. The repository contains the raw motor runs and analysis, geometry and mesh-generation scripts, design-gradient sweeps and timing records, and solver verification logs. Candidate-validation records are pinned at `224332c5`, and independent reference reruns at `8acd21ec`. Reference data are extracted from the authors' ICEM snapshot at `b145aaf4`, a later repository revision rather than a claim to the original conference release. Source hashes, extraction checks and the upstream GPL-3.0-or-later license are retained. The paper package includes a manifest, copied figure inputs and a regeneration script; it does not require rerunning FEM to rebuild the figures. Recomputing the physical study requires the separately documented mesh and solver commands; the benchmark READMEs record the GEODE-side Gmsh and Rust toolchain versions.

A Linear slotless screening control

The stored slotless model has 24 distinct angles. Fitting $T(\phi) = c_0 + c_1 \cos\phi + c_2 \sin\phi$, with electrical angle $\phi = p\theta_r$, mechanical rotor angle θ_r and pole-pair count p, on alternate angles and evaluating on the other twelve gives normalized held-out RMSE 3.08×10^{-10} and maximum error 6.82×10^{-10}, using the recorded analytic-amplitude normalization. This is a locked-rotor angle sweep, not synchronous operating ripple. For a fixed linear field operator and sinusoidal sources, the field lies in a small basis of source responses; a quadratic torque functional can contain only the corresponding finite set of harmonics. The particular model needs just the three fitted coefficients. This is an angular interpolation test at one geometry, not a shape-gradient test or evidence of generalization across designs.

B Design points for derivative checks

Table 4 identifies the scalar derivative-study coordinates and the saved Opt120 coordinates. Values are rounded for display; the accompanying records retain full precision. The operating angle is stored in degrees and converted to radians in the current equation. These coordinates describe the scalar design space; the saved reference optimum additionally contains surface-control offsets.

Table 4: Initial and interior perturbed design points for derivative checks, alongside Opt120 for the forward validation. Lengths refer to the unscaled geometry; surface-control offsets are retained in the data.

Parameter	Meaning	Unit	Initial	Perturbed	Opt120
MAG	Pole-axis iron depth above magnet	mm	7.0000	6.5506	6.1011
MH	Radial magnet height	mm	7.0000	4.8722	2.7444
MW	Magnet width	mm	19.0000	19.8670	20.7340
SW1	Stator tooth width	mm	4.0000	4.0309	4.0619
SW2	Slot-opening width	mm	2.3000	1.6500	1.0001
SW3	Slot-opening depth	mm	1.0000	0.7502	0.5004
SW4	Back-iron depth on slot axis	mm	8.2500	6.8418	5.4336
SD1	Stator outer diameter	mm	135.0000	132.6466	130.2931
Sr1	Slot-bottom fillet radius	mm	1.0000	1.4999	1.9998
LENGTH	Axial length	mm	100.0000	90.0003	80.0005
ϕ_0	Electrical current offset	°	0.0000	-5.8231	-11.6463
ScaleR	Radial similarity scale	1	1.0000	1.0239	1.0477

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